Lesson Planning

Complete Lesson Plan Book

Class: FSc Pre-Medical · Class 11

Subject: Physics

Medium: English

Academic Year 2026-27

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This book is prepared as a teaching guide for teachers.

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Unit 1: Newton's Laws of Motion

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Learning Objectives

• Students will state Newton's three laws of motion in their own words.

• Students will give one everyday example of each law.

• Students will explain the link between force, mass and acceleration using the second law.

Key Concepts

• Newton's first law says a body stays at rest or in uniform motion unless an unbalanced force acts on it — this is the law of inertia.

• A passenger jerks forward when a moving bus stops suddenly because the body tends to keep moving — an everyday example of inertia.

• Newton's second law says force equals mass times acceleration, F = ma, so a heavier object needs a bigger force for the same acceleration.

• Newton's third law says every action has an equal and opposite reaction, as when a rocket pushes gas down and the gas pushes the rocket up.

• Friction, weight and the normal reaction are common forces that appear in examples of all three laws.

Teaching Methodology (Step by Step)

1. Roll a ball across the desk and ask why it finally stops, to open the idea of force and motion.
2. State the first law and explain inertia with the bus-passenger example.
3. State the second law, write F = ma on the board, and work out one simple numerical example.
4. State the third law and show action-reaction pairs with a balloon rocket and a swimmer pushing off a wall.
5. Ask students to match five everyday situations to the law each one shows.
6. Summarise all three laws and ask students to repeat each law with its example.

Classroom Activities

• Coin and card trick: flick a card from under a coin to show inertia in action.

• Force and mass race: push a light and a heavy book with the same effort and compare how fast each moves.

• Action-reaction hunt: students list three action-reaction pairs they can see in the classroom.

Assessment Questions

1. State Newton's first law and give one example from daily life.
2. A force of 10 N acts on a mass of 2 kg. What is the acceleration?
3. When you jump off a boat, the boat moves backwards. Which law explains this, and why?

Homework

• Write one example of each of Newton's laws that you observed at home today.

• Solve: a 5 kg trolley is pushed with a force of 15 N. Find its acceleration, and find the force needed to double it.

Time Allocation

40 minutes